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CARRY CASE AND RECORDING SYSTEM

Abstract

The carry case includes a housing main body; a first accommodation section configured to accommodate a plurality of recording devices in a state of being stacked in plurality of tiers in a vertical direction in the first accommodation region within the housing main body; a second accommodation section configured to accommodate the power supply device that supplies electric power to the plurality of recording devices in the second accommodation region within the housing main body; a first opening and closing door configured to open and close in a first surface; a second opening and closing door configured to open and close on a second surface; and a wheel configured to move the housing main body.

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Background/Summary

[0001] The present application is based on, and claims priority from JP Application Serial Number 2024-032759, filed Mar. 5, 2024, and from JP Application Serial Number 2024-032760, filed Mar. 5, 2024, the disclosures of which are hereby incorporated by reference herein in their entirety.

BACKGROUND

1. Technical Field

[0002] The present disclosure relates to a carry case and a recording system.

2. Related Art

[0003] For example, JP-A-2021-87130 discloses a cabinet capable of accommodating a recording device that performs recording by ejecting liquid. A reading device and a projection device are installed in the cabinet, and it is possible to improve convenience for the user.

[0004] However, although the cabinet can accommodate the recording device, it is not easy to transport the cabinet with the recording device inside. As described above, it is desired to improve user convenience.

SUMMARY

[0005] A carry case for solving the above problem includes a housing main body; a first accommodation section configured to accommodate, in a first accommodation region within the housing main body, a plurality of recording devices in a state of being stacked in a plurality of tiers in a vertical direction; a second accommodation section configured to accommodate, in the housing main body in a second accommodation region that is aligned in the vertical direction with the first accommodation region, a power supply device that supplies electric power to the plurality of recording devices; a first opening and closing door configured to open and close in a first surface of the housing main body; a second opening and closing door configured to open and close in a second surface opposite to the first surface of the housing main body; and a wheel that is below the housing main body and that enables moving the housing main body.

[0006] A carry case for solving the above problem includes a housing main body; a first accommodation section configured to accommodate a plurality of recording devices in a state of being stacked in a plurality of tiers in a vertical direction in a first accommodation region within the housing main body, the recording devices performing recording by ejecting liquid; a second accommodation section configured to accommodate, in the housing main body in a second accommodation region that is aligned in the vertical direction with the first accommodation region, a power supply device that supplies electric power to the plurality of recording devices; a wheel that is below the housing main body and that enables moving the housing main body; and a horizontal maintaining section configured to, when the housing main body is inclined at a predetermined angle, horizontally support the plurality of recording devices accommodated in the first accommodation section.

[0007] A recording system for solving the above problem includes a plurality of recording devices; a power supply device configured to supply power to the plurality of recording devices; and a carry case configured to accommodate the plurality of recording devices and the power supply device, wherein the carry case has a housing main body, a first accommodation section configured to accommodate the plurality of recording devices in a state of being stacked in a plurality of tiers in a vertical direction in the first accommodation region in the housing main body, a second accommodation section configured to house the power supply device in a second accommodation region aligned with the first accommodation region in the vertical direction in the housing main

body, a first opening and closing door configured to open and close in a first surface of the housing main body, a second opening and closing door configured to open and close in a second surface opposite to the first surface of the housing main body, and a wheel that is below the housing main body and that enables moving the housing main body.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. 1 is a perspective view showing a recording system according to the first embodiment.

[0009] FIG. 2 is a perspective view showing a recording system of the first embodiment.

[0010] FIG. 3 is a schematic diagram showing a recording system of the first embodiment.

[0011] FIG. 4 is a schematic diagram showing a recording system of the first embodiment.

[0012] FIG. 5 is a schematic diagram showing a recording system of the first embodiment.

[0013] FIG. 6 is a perspective view showing a recording device of the first embodiment.

[0014] FIG. 7 is a perspective view illustrating a recording device of the first embodiment.

[0015] FIG. 8 is a perspective view showing a recording device of the first embodiment.

[0016] FIG. 9 is a cross-sectional view showing a recording device of the first embodiment.

DESCRIPTION OF EMBODIMENTS

First Embodiment

[0017] Hereinafter, an embodiment of a recording system including a carry case will be described.

In the following description, a direction intersecting the vertical direction Z is referred to as a widthwise direction X, and a direction intersecting the vertical direction Z and the widthwise direction X is referred to as a front-rear direction Y. One direction along the widthwise direction X is defined as the first widthwise direction X1, and the other direction along the widthwise direction X is defined as the second widthwise direction X2. One direction along the front-rear direction Y is referred to as the frontward direction Y1, and the other direction along the front-rear direction Y is referred to as the backward direction Y2. In the vertical direction Z, upward is referred to as the upward direction Z1, and downward is referred to as the downward direction Z2.

Configuration of Recording System 10

[0018] As shown in FIG. 1, a recording system 10 includes a carry case 100, a recording device 11, and a power supply device 91. The recording system 10 may include a plurality of recording devices 11. The recording system 10 may include a first recording device 11A, a second recording device 11B, and a third recording device 11C as the plurality of recording devices 11.

[0019] The plurality of recording devices 11 can be stacked in the plurality of tiers in the vertical direction Z. The plurality of recording devices 11 can be accommodated in the carry case 100 in a state of being stacked in the plurality of tiers in the vertical direction Z. The second recording device 11B is stacked in the upward direction Z1 of the first recording device 11A. The third recording device 11C is stacked in the upward direction Z1 of the second recording device 11B. The upper limit of the plurality of tiers is, for example, three tiers, but is not limited thereto.

[0020] The carry case 100 can accommodate the recording device 11 and the power supply device 91. The carry case 100 can accommodate the plurality of recording devices 11 and the power supply device 91. The carry case 100 can be carried in a state in which the plurality of recording devices 11 and the power supply device 91 are accommodated. The recording device 11 may be accommodated in the carry case 100 such that a front surface 12 faces the frontward direction Y1 and a rear surface 13 faces the backward direction Y2.

[0021] The recording device 11 is configured to record an image on a medium. The recording device 11 may be configured to record an image on a medium by ejecting a liquid onto the medium. The liquid, for example, may be ink. That is, the recording device 11 may be an inkjet type printer. The liquids may have, for example, a plurality of colors type or just mono color type.

The medium may be, for example, paper sheet. The recording device **11** may be a device capable of performing recording on a plurality of types of paper sheets.

[0022] The power supply device **91** supplies power to the recording device **11**. The power supply device **91** supplies power to the plurality of recording devices **11**. The power supply device **91** is configured to be portable. The power supply device **91** may be a portable power source. The power supply device **91** can be accommodated in the carry case **100**.

Configuration of Carry Case **100**

[0023] The carry case **100** may be configured to be in the shape of a rectangular parallelepiped. The carry case **100** is constructed in a rectangular parallelepiped shape that is longer in the front-rear direction Y than in the widthwise direction X. However, it is not limited to this configuration, and, for example, it may also be constructed in a rectangular parallelepiped shape that is longer in the widthwise direction X than in the front-rear direction Y.

[0024] The carry case **100** may include a case front surface **100A**, a case rear surface **100B**, a case left side surface **100C**, a case right side surface **100D**, a case bottom surface **100E**, and a case top surface **100F**. The case front surface **100A** corresponds to an example of a first surface. The case rear surface **100B** corresponds to an example of a second surface.

[0025] The carry case **100** includes a housing main body **60**, a first opening and closing door **61**, a second opening and closing door **62**, wheels **63**, a grip section **64**, and a table **65**. The housing main body **60** may be constructed in a rectangular parallelepiped shape.

[0026] The housing main body **60** includes a first opening section **60A** and a second opening section **60B**. The first opening section **60A** is provided in the frontward direction Y1. The first opening section **60A** can be opened and be closed by the first opening and closing door **61**. The second opening section **60B** is provided in the backward direction Y2. The second opening section **60B** can be opened and be closed by the second opening and closing door **62**.

Configuration of First Accommodation Section **71** and Second Accommodation Section **72**

[0027] The housing main body **60** includes a first accommodation section **71** and a second accommodation section **72**. That is, the carry case **100** includes the first accommodation section **71** and the second accommodation section **72**. The first accommodation section **71** and the second accommodation section **72** are provided in the housing main body **60**. The first accommodation section **71** and the second accommodation section **72** are positioned to be aligned in the vertical direction Z. The first accommodation section **71** may be positioned in the upward direction Z1 of the second accommodation section **72**.

[0028] The first accommodation section **71** can accommodate the plurality of recording devices **11** in a state of being stacked in plurality of tiers in the vertical direction Z. The first accommodation section **71** includes a first placement section **71A** and a first accommodation region **71B**. The first placement section **71A** is provided within the housing main body **60**. The recording device **11** can be placed on the first placement section **71A**. The first placement section **71A** can support the plurality of recording devices **11** in a stacked state in plurality of tiers in the vertical direction Z. The first placement section **71A** may have a flat-plate shape.

[0029] The first accommodation region **71B** is provided within the housing main body **60**. The first accommodation region **71B** is a region where the recording device **11** can be accommodated. The first accommodation region **71B** is the region where the plurality of recording devices **11** can be accommodated in a state of being stacked in the plurality of tiers in the vertical direction z.

[0030] The second accommodation section **72** can accommodate the power supply device **91** in the housing main body **60**. The second accommodation section **72** includes a second placement section **72A** and a second accommodation region **72B**. The second placement section **72A** is provided within the housing main body **60**. The power supply device **91** can be placed on the second placement section **72A**. The second placement section **72A** may have a flat-plate shape. The second placement section **72A** may be a bottom plate of the housing main body **60**.

[0031] The second accommodation region **72B** is provided within the housing main body **60**. The

second accommodation region **72B** is a region aligned with the first accommodation region **71B** in the vertical direction **Z**. The second accommodation region **72B** is a region capable of housing the power supply device **91**.

Configuration of First Opening and Closing Door **61** and Second Opening and Closing Door **62**

[0032] The first opening and closing door **61** is provided in the frontward direction **Y1** of the housing main body **60**. The first opening and closing door **61** may constitute the case front surface **100A**. The first opening and closing door **61** can open and close the first opening section **60A** in the case front surface **100A**. The first opening and closing door **61** may be rotatable about a first rotation shaft along the vertical direction **Z**. The first rotation shaft may be provided at the second widthwise direction **X2** side of the carry case **100**.

[0033] When the first opening and closing door **61** is opened, the first accommodation section **71** and the second accommodation section **72** can be accessed from the frontward direction **Y1**. That is, the first opening and closing door **61** can open and close the first accommodation region **71B** and the second accommodation region **72B** in the frontward direction **Y1**.

[0034] The second opening and closing door **62** is provided in the backward direction **Y2** of the housing main body **60**. The second opening and closing door **62** may constitute the case rear surface **100B**. The second opening and closing door **62** can open and close the second opening section **60B** in the case rear surface **100B**. The second opening and closing door **62** may be rotatable about a second rotation shaft along the vertical direction **Z**. The second rotation shaft may be provided on the second widthwise direction **X2** side of the carry case **100**.

[0035] When the second opening and closing door **62** is opened, the first accommodation section **71** and the second accommodation section **72** can be accessed from the backward direction **Y2**. That is, the second opening and closing door **62** can open and close the first accommodation region **71B** and the second accommodation region **72B** in the backward direction **Y2**.

[0036] The wheels **63** are provided in the downward direction **Z2** from the housing main body **60**. The wheels **63** can be placed on an installation surface **99**. When the carry case **100** is arranged horizontally, all of the wheels **63** are arranged on the installation surface **99**. When the carry case **100** is positioned so as to be inclined, some of the wheels **63** are positioned on the installation surface **99**.

[0037] The wheels **63** enable the housing main body **60** to move. That is, the wheels **63** allow the carry case **100** to be movable. The wheels **63** may move the carry case **100** by a driving force from a drive section **69** (to be described later).

[0038] The grip section **64** can be gripped by a user. The grip section **64** can be gripped when the carry case **100** is handled by the user. The grip section **64** is located on the backward direction **Y2** side of the housing main body **60**. The grip section **64** is located on the upward direction **Z1** of the housing main body **60**. The grip section **64** may be expanded and contracted so it can be housed in the housing main body **60**. In particular, when the user moves the carry case **100**, the grip section **64** can be gripped in a state where the grip section **64** is extended to the upward direction **Z1** and the carry case **100** is inclined to the backward direction **Y2**.

[0039] As shown in FIGS. **1** and **2**, the table **65** is provided in the upward direction **Z1** of the housing main body **60**. The table **65** can be displaced between a storage position **P11** shown in FIG. **1** and an expansion position **P12** shown in FIG. **2**. In other words, the table **65** is deployable from the storage position **P11** to the expansion position **P12**. The table **65** may be displaceable between the storage position **P11** and the expansion position **P12** by rotating about a third rotation shaft that extends along the front-rear direction **Y** on the second widthwise direction **X2** side of the housing main body **60**.

[0040] As shown in FIG. **1**, the storage position **P11** is a position at which table **65** is disposed in the upward direction **Z1** of the housing main body **60**. The storage position **P11** is a position at which a first placement surface **65A** of the table **65** is exposed in the upward direction **Z1**. The first placement surface **65A** may constitute the case top surface **100F** when arranged at the storage

position **P11**.

[0041] An object to be placed can be placed on the first placement surface **65A**. The object to be placed may be, for example, a laptop computer, a tablet terminal device, or a portable terminal device. The object to be placed may be capable of communicating with the recording device **11**. The object to be placed may be supplied with power from the power supply device **91**.

[0042] As shown in FIG. **2**, the expansion position **P12** is a position at which the table **65** is disposed further in the widthwise direction **X** than the housing main body **60**. The expansion position **P12** is a position at which a second placement surface **65B** of the table **65** is exposed to the upward direction **Z1**. The second placement surface **65B** is the back surface of the first placement surface **65A**. An object to be placed can be placed on the second placement surface **65B**. That is, when the table **65** is expanded to the expansion position **P12**, the object to be placed can be placed on the second placement surface **65B**. The second placement surface **65B** corresponds to an example of a placement surface. The expansion position **P12** is a position at which a housing top surface **60C** of the housing main body **60** is exposed to the upward direction **Z1**. An object to be placed can be placed on the housing top surface **60C**.

[0043] When the table **65** is expanded to the expansion position **P12**, the table **65** is supported by the opened first opening and closing door **61** and the second opening and closing door **62**. In detail, when the table **65** is extended to the expansion position **P12**, the table **65** is supported by a first top end section **61A** of the opened first opening and closing door **61** and a second top end section **62A** of the opened second opening and closing door **62**.

[0044] The carry case **100** may include an adjuster **66**. The adjuster **66** is provided in the downward direction **Z2** from the first opening and closing door **61**. The adjuster **66** may be provided in the downward direction **Z2** from the first opening and closing door **61** on the first widthwise direction **X1** side when the first opening and closing door **61** is closed. In this way, the adjuster **66** may be provided on the side opposite from the first rotation shaft of the first opening and closing door **61** in the widthwise direction **X**.

[0045] The adjuster **66** is provided in the downward direction **z2** from the second opening and closing door **62**. The adjuster **66** may be provided in the downward direction **Z2** on the first widthwise direction **X1** side of the second opening and closing door **62** when the second opening and closing door **62** is closed. In this way, the adjuster **66** may be provided on the side opposite to the second rotation shaft of the second opening and closing door **62** in the widthwise direction **X**.

[0046] The height of the adjuster **66** in the vertical direction **Z** can be adjusted. Thus, when the table **65** is expanded to the expansion position **P12**, the height of the adjuster **66** can be adjusted in the downward direction **22** of the first opening and closing door **61**. The adjuster **66** is configured to support the first opening and closing door **61** from the downward direction **22** by adjusting the height when the table **65** is expanded to the expansion position **P12**.

[0047] When the table **65** is expanded to the expansion position **P12**, the height of the adjuster **66** can be adjusted in the downward direction **22** of the second opening and closing door **62**. The adjuster **66** is configured to support the second opening and closing door **62** from the downward direction **22** by adjusting the height when the table **65** is expanded to the expansion position **P12**.

[0048] As shown in FIG. **3**, the carry case **100** may include a first locking section **67A** and a second locking section **67B**. The first locking section **67A** is provided in the housing main body **60** and the first opening and closing door **61**. The first locking section **67A** locks the first opening and closing door **61** in a state where the first opening and closing door **61** is closed. The second locking section **67B** is provided on the housing main body **60** and the second opening and closing door **62**. The second locking section **67B** locks the second opening and closing door **62** in a state where the second opening and closing door **62** is closed.

[0049] The carry case **100** may include an illumination section **68**. The illumination section **68** is provided in the housing main body **60**. The illumination section **68** illuminates the inside of the housing main body **60**. The illumination section **68** may illuminate the first accommodation region

71B. The illumination section **68** may illuminate the second accommodation region **72B**.

[0050] The carry case **100** may include the drive section **69**. The drive section **69** can be housed in the second accommodation section **72**. The drive section **69** is fixed to the second accommodation section **72**. The drive section **69** is configured to drive the wheels **63**. The drive section **69** is a driving source that drives the wheels **63** by supplying a driving force to the wheels **63**. The drive section **69** may be a driving source that drives the wheels **63** located in the backward direction **Y2**. The drive section **69** may be a motor.

[0051] The carry case **100** may include a first fixing section **70**. The first fixing section **70** is provided on the second placement section **72A**. That is, the second accommodation section **72** may include the first fixing section **70**. Even when the housing main body **60** is inclined at a predetermined angle, the first fixing section **70** fixes the position of the power supply device **91** placed on the second placement section **72A**.

[0052] The first fixing section **70** may include a first member **70A** that fixes the position of the power supply device **91** placed on the second placement section **72A** in the widthwise direction **X**. The first member **70A** is displaceable in the widthwise direction **X** in accordance with the dimension of the power supply device **91** in the widthwise direction **X**. The first member **70A** can be fixed at an arbitrary position to be displaced in the widthwise direction **X**.

[0053] The first fixing section **70** may include a second member **70B** that fixes the position of the power supply device **91** placed on the second placement section **72A** in the front-rear direction **Y**. The second member **70B** is displaceable in the front-rear direction **Y** in accordance with the dimension of the power supply device **91** in the front-rear direction **Y**. The second member **70B** can be fixed at any position where it is displaced in the front-rear direction **Y**.

Configuration of Horizontal Maintaining Section **73**

[0054] The carry case **100** includes a horizontal maintaining section **73**. The horizontal maintaining section **73** is configured to maintain horizontal the plurality of recording devices **11** accommodated in the first accommodation section **71** when the housing main body **60** is inclined at a predetermined angle. The predetermined angle may be, for example, an angle larger than 0 degrees and smaller than 30 degrees with respect to the backward direction **Y2**. That is, the predetermined angle may be an angle inclined with respect to the backward direction **Y2**. The horizontal maintaining section **73** does not horizontally maintain the power supply device **91** accommodated in the second accommodation section **72** when the housing main body **60** is inclined at a predetermined angle.

[0055] The horizontal maintaining section **73** includes a guide section **74**. The guide section **74** is fixed to the housing main body **60**. The guide section **74** may have a circular arc shape about an axis along the widthwise direction **X**. When the carry case **100** is arranged horizontally, the guide section **74** may extend to the same height as the top surface of the first placement section **71A** in the frontward direction **Y1**. In this manner, the guide section **74** is extended to a position lower than the plurality of recording devices **11** stacked on the first placement section **71A** in the frontward direction **Y1**.

[0056] In a case where the carry case **100** is horizontally arranged, the guide section **74** may extend to a position higher than the top surface of the first placement section **71A** on the backward direction **Y2**. In this manner, the guide section **74** is extended to a position higher than the plurality of recording devices **11** stacked on the first placement section **71A** in the backward direction **Y2**. The guide section **74** guides the first placement section **71A** so as to maintain the first placement section **71A** horizontally.

[0057] The guide section **74** may include a guide groove **74A**. The guide groove **74A** is provided on the inner peripheral side of the guide section **74**. The guide groove **74A** is slidable on a rotator **71C** provided on the first placement section **71A**. The rotator **71C** may be, for example, a roller. The guide groove **74A** is provided at a position corresponding to the movement range of the first placement section **71A**. In this manner, the guide section **74** guides the first placement section **71A**

so as to maintain the horizontal of the first placement section **71A** by rotating the first placement section **71A** about an axis along the widthwise direction **X**.

[0058] An outer peripheral section **74B** of the guide section **74** is slidable with a sliding portion **71D** provided on the first placement section **71A**. For this reason, even when the carry case **100** receives vibration in the vertical direction **Z**, the sliding portion **71D** comes into contact with the outer peripheral section **74B**, and thus the first placement section **71A** can maintain a certain distance without being separated from the guide section **74**. That is, it is configured so that the distance between the guide section **74** and the first placement section **71A** is maintained within the permissible range. The permissible range may be a range in which the first placement section **71A** is not separated from the guide section **74** and is not affected by the vibration.

[0059] The horizontal maintaining section **73** may include a second fixing section **75**. That is, the carry case **100** may include the second fixing section **75**. The second fixing section **75** is arranged in the housing main body **60**. In a case where the housing main body **60** is arranged horizontally, the second fixing section **75** is configured to be insertable and removable by a user's operation on the bottom surface of the first placement section **71A**.

[0060] When the housing main body **60** is arranged horizontally, the second fixing section **75** is arranged so as to support the bottom surface of the first placement section **71A**, thereby fixing the first placement section **71A** horizontally. By this, in a case where the housing main body **60** is arranged horizontally, the second fixing section **75** fixes the plurality of recording devices **11** accommodated in the first accommodation section **71** so as to be maintained horizontally. When the housing main body **60** is arranged horizontally, the second fixing section **75** is arranged so as not to support the bottom surface of the first placement section **71A**, thereby releasing the fixing of the first placement section **71A**.

Posture of Carry Case **100**

[0061] As shown in FIG. **4**, in a case where the second fixing section **75** horizontally fixes the first placement section **71A** in a state where the housing main body **60** is horizontally arranged, the plurality of recording devices **11** accommodated in the first accommodation section **71** are horizontally fixed. By this, it is possible to perform recording with the recording device **11** in a stable state.

[0062] As shown in FIG. **5**, after the fixation of the first placement section **71A** by the second fixing section **75** is released, when the housing main body **60** is inclined during the transport of the carry case **100**, the first placement section **71A** is guided by the guide section **74** so as to be maintained horizontal. By this, the horizontal maintaining section **73** can horizontally maintain the plurality of recording devices **11** accommodated in the first accommodation section **71**.

[0063] The power supply device **91** is fixed to the second placement section **72A** by the first fixing section **70**. By this, even if the power supply device **91** is not maintained horizontally, the power supply device **91** is stably accommodated in the second accommodation section **72**.

Configuration of Recording Device **11**

[0064] As shown in FIG. **6**, the recording device **11** is constructed in a rectangular parallelepiped shape. The recording device **11** has the front surface **12**. The front surface **12** is a plane facing the frontward direction **Y1**. The recording device **11** has the rear surface **13**. The rear surface **13** is a flat surface facing the backward direction **Y2**. The recording device **11** has a left side surface **14**. The left side surface **14** is a flat surface facing the second widthwise direction **X2**. The recording device **11** has a right side surface **15**. The right side surface **15** is a flat surface facing the first widthwise direction **X1**. The recording device **11** has a bottom surface **16**. The bottom surface **16** is a flat surface facing the downward direction **Z2**. The recording device **11** has a top surface **17**. The top surface **17** is a flat surface facing the upward direction **Z1**.

[0065] As described above, the top surface **17** includes a flat surface on which the placed object can be placed. The placed object may be another recording device **11** or may be a laptop computer. The top surface **17** may be entirely flat in the region or may include non-flat areas in the region,

provided that there is a flat portion where the placed object can be placed.

[0066] The recording device **11** includes a housing **20** and a main body **21**. The housing **20** is configured to accommodate the main body **21**. The housing **20** constitutes at least a part of the left side surface **14**, a part of the right side surface **15**, a part of the bottom surface **16**, and a part of the top surface **17** of the recording device **11**. However, the housing **20** may also constitute at least the entire left side surface **14**, the entire right side surface **15**, the entire bottom surface **16**, and the entire top surface **17** of the recording device **11**.

[0067] As described above, the housing **20** has a plurality of surfaces such as the left side surface **14**, the right side surface **15**, the bottom surface **16**, and the top surface **17** of the recording device **11**. It can be said that the left side surface **14**, the right side surface **15**, the bottom surface **16**, and the top surface **17** of the recording device **11** are the left side surface, the right side surface, the bottom surface, and the top surface of the housing **20**, respectively. At least a part of the front surface **12** of the recording device **11** may be included in the housing **20**. Further, at least a part of the rear surface **13** of the recording device **11** may be included in the housing **20**.

[0068] The main body **21** constitutes at least the entire front surface **12** of the recording device **11** but may constitute at least a part of the front surface **12** of the recording device **11**. Thus, the main body **21** has the front surface **12** of the recording device **11**. It can be said that the front surface **12** of the recording device **11** is the front surface of the main body **21**. In addition, the rear surface **13** of the recording device **11** may be constituted by the rear surface of the housing **20** and the rear surface of the main body **21**, may be constituted by only the rear surface of the housing **20**, or may be constituted by only the rear surface of the main body **21**.

[0069] The recording device **11** includes a contained-amount visual check section **31**. The contained-amount visual check section **31** is provided in the main body **21**. The contained-amount visual check section **31** can be used to visually check a contained amount of liquid in a liquid container **40** (to be described later). The contained-amount visual check section **31** includes a plurality of visual check windows respectively corresponding to the liquid of the respective colors.

[0070] The recording device **11** includes an operation panel **33**. The operation panel **33** is provided on the main body **21**. The operation panel **33** includes an operation section **34** and a display section **35**. The operation section **34** can be operated by a user. The operation section **34** may include a plurality of operation buttons. The display section **35** displays information related to the recording device **11**. The operation panel **33** is positioned on the front surface **12** of the recording device **11**.

[0071] The main body **21** is configured to be displaceable along the front-rear direction Y with respect to the housing **20**. In detail, the main body **21** is configured to be displaceable between a closed position, a first open position, and a second open position along the front-rear direction Y with respect to the housing **20**.

[0072] The main body **21** can be positioned at a closed position. The closed position is a position at which the main body **21** is accommodated in the housing **20**. The closed position is a position at which the main body **21** protrudes from the housing **20** by a predetermined distance DO toward the frontward direction Y1. However, the main body **21** may be a position at which it does not protrude from the housing **20**. That is, the main body **21** constitutes a part of the left side surface **14**, a part of the right side surface **15**, a part of the bottom surface **16**, and a part of the top surface **17** of the recording device **11**. However, the main body **21** may not constitute the entire left side surface **14**, the right side surface **15**, the bottom surface **16**, and the top surface **17** of the recording device **11**.

[0073] As shown in FIG. 7, the main body **21** can be displaced to the first open position. The first open position is an open position at which the main body **21** protrudes from the housing **20** toward the frontward direction Y1. In particular, the first open position is a position at which the main body **21** protrudes from the housing **20** toward the frontward direction Y1 by a first distance D1. The first distance D1 is longer than the predetermined distance DO. In this way, when the main body **21** is arranged at the first open position, at least a part of the interior of the main body **21** is exposed from the housing **20**.

[0074] As shown in FIG. 8, the main body **21** can be displaced to the second open position. The second open position is an open position at which the main body **21** protrudes from the housing **20** toward the frontward direction **Y1**. In particular, the second open position is a position at which the main body **21** protrudes from the housing **20** toward the frontward direction **Y1** by a second distance **D2**. The second distance **D2** is longer than the first distance **D1**. In this way, when the main body **21** is arranged at the second open position, at least a part of the interior of the main body **21** is exposed from the housing **20**.

[0075] The recording device **11** includes a control section **23**. The control section **23** is mounted on the main body **21**. The control section **23** is housed in the housing **20**. The control section **23** is not exposed from the housing **20** when the main body **21** is positioned at the closed position, the first open position, and the second open position.

[0076] The control section **23** controls the recording device **11**. The control section **23** controls various operations performed by the recording device **11**. The control section **23** can be configured as a circuit including a: one or more processors that execute various processes in accordance with a computer program, B: one or more dedicated hardware circuits that execute at least some of the various processes, or y: a combination thereof. The hardware circuit is, for example, an application-specific integrated circuit. The processor includes a CPU and memory, such as RAM and ROM, and the memory stores program code or commands configured to cause the CPU to perform processes. The memory, in other words, computer-readable medium, includes any readable medium that can be accessed by a general purpose or special purpose computer.

[0077] The recording device **11** includes an opening and closing section **37**. The opening and closing section **37** is arranged in the main body **21**. The opening and closing section **37** is positioned on the front surface **12** of the recording device **11**. The opening and closing section **37** is configured to be rotatable around a rotation shaft along the widthwise direction **X** at a lower end portion **37A**. By this, the opening and closing section **37** is configured to open and close with respect to the front surface **12** of the recording device **11**.

[0078] The recording device **11** includes a discharge port **38**. The discharge port **38** is provided in the main body **21**. The discharge port **38** is an opening for discharging a medium on which recording is performed by a recording section **48** (to be described later). The discharge port **38** is opened in the frontward direction **Y1** of the recording device **11**. That is, the discharge port **38** is provided so as to be opened to the frontward direction **Y1** in the front surface **12** of the recording device **11**. In other words, the discharge port **38** is provided on the front surface of the main body **21** so as to open to the frontward direction **Y1**.

[0079] The discharge port **38** is exposed in the frontward direction **Y1** of the recording device **11** when the opening and closing section **37** is open. The discharge port **38** is not exposed when the opening and closing section **37** is closed. However, the discharge port **38** may be exposed even when the opening and closing section **37** is closed. The discharge port **38** is positioned in the second widthwise direction **X2** of the liquid container **40** (to be described later).

[0080] The recording device **11** includes a sheet discharge tray **37B**. The sheet discharge tray **37B** is provided in the main body **21**. The sheet discharge tray **37B** is provided on an inner wall surface of the opening and closing section **37**. The sheet discharge tray **37B** is exposed when the opening and closing section **37** is opened, and a medium after recording can be placed thereon. The sheet discharge tray **37B** may be configured to expand and contract in the front-rear direction **Y**.

[0081] The recording device **11** includes a device-grip section **39**. The device-grip section **39** is provided in the main body **21**. The device-grip section **39** is provided in the upward direction **Z1** of the discharge port **38**. The device-grip section **39** is provided at a position on the front surface **12** of the recording device **11** where it can be gripped by the user from the frontward direction **Y1**. That is, the device-grip section **39** is provided at a position where the user can grip the grip section from the frontward direction **Y1** on the side of the surface facing the frontward direction **Y1** of the main body **21**. In this way, the device-grip section **39** can be gripped by the user when the main body **21**

is displaced in the front-rear direction Y. In particular, even when the plurality of recording devices **11** are stacked in the vertical direction Z or the plurality of recording devices **11** are installed side by side in the widthwise direction X, the device-grip section **39** can be easily gripped by the user from the frontward direction Y1 side.

[0082] As shown in FIG. 7, the recording device **11** includes the liquid container **40**. The liquid container **40** is mounted on the main body **21**. The liquid container **40** is configured to contain liquid used for recording on a medium. In other words, the liquid container **40** is configured to contain a liquid to be supplied to the recording section **48** (to be described later). The liquid container **40** may be a tank-type container into which liquid can be poured or may be a cartridge-type container which can be attached to and detached from the recording device **11**.

[0083] The liquid container **40** includes an accommodation chamber **41**, a liquid inlet **42**, and a cap **43**. The accommodation chamber **41** is configured to accommodate a liquid. The liquid inlet **42** is an opening through which liquid can be poured into the accommodation chamber **41**. The cap **43** covers the liquid inlet **42**. The cap **43** can suppress drying in the accommodation chamber **41** by sealing the accommodation chamber **41**. By removing the cap **43** from the liquid inlet **42**, the liquid can be poured via the liquid inlet **42** into the accommodation chamber **41**. The liquid container **40** may include a plurality of accommodation chambers **41**, a plurality of liquid inlets **42**, and a plurality of caps **43** corresponding to the respective colors.

[0084] The liquid container **40** is not exposed from the housing **20** when the main body **21** is positioned at the closed position. The liquid container **40** is exposed from the housing **20** when the main body **21** is positioned at the first open position. In this way, when the main body **21** is positioned at the first open position, by removing the cap **43**, the liquid inlet **42** is exposed from the housing **20**. By this, the liquid can be poured via the liquid inlet **42** into the accommodation chamber **41**. The liquid container **40** is also exposed from the housing **20** when the main body **21** is positioned at the second open position.

[0085] The recording device **11** includes a waste liquid mount section **45**. The waste liquid mount section **45** is configured to allow a waste liquid container **46** to be mounted. The waste liquid container **46** is configured to be able to collect liquid discharged as waste liquid from the recording section **48** (to be described later) via a maintenance device (not shown). The waste liquid mount section **45** is mounted on the main body **21**. The waste liquid mount section **45** is provided in the upward direction Z1 of the discharge port **38**.

[0086] The waste liquid mount section **45** is not exposed from the housing **20** when the main body **21** is disposed at the closed position. The waste liquid mount section **45** is exposed from the housing **20** when the main body **21** is positioned at the first open position. In this way, when the main body **21** is disposed at the first open position, the waste liquid mount section **45** is exposed from the housing **20**. By this, the waste liquid container **46** can be attached to and detached from the waste liquid mount section **45**. The waste liquid mount section **45** is also exposed from the housing **20** when the main body **21** is positioned at the second open position.

[0087] As shown in FIG. 8, the recording device **11** includes the recording section **48**. The recording section **48** is configured to perform recording on a medium. The recording section **48** may be configured to perform recording on a medium by ejecting liquid onto the medium. The recording section **48** is mounted on the main body **21**. The recording section **48** is provided at a position adjacent to the liquid container **40** in the front-rear direction Y. The recording section **48** is provided in the backward direction Y2 of the liquid container **40**.

[0088] The recording section **48** is exposed from the housing **20** when the main body **21** is positioned at the second open position. In particular, a carriage **51** (to be described later) is exposed from the housing **20** when the main body **21** is positioned at the second open position. The recording section **48** is not exposed from the housing **20** when the main body **21** is positioned at the closed position and the first open position.

[0089] The recording device **11** includes a transport path **49**. The transport path **49** is a path for

transporting a medium. The transport path **49** is provided in the main body **21**. A part of the transport path **49** is exposed from the housing **20** when the main body **21** is positioned at the second open position. In particular, in the transport path **49**, a recording region RA where the recording section **48** performs recording on the medium is exposed from the housing **20** when the main body **21** is positioned at the second open position. A portion of the transport path **49** is not exposed from the housing **20** when the main body **21** is positioned at the closed position and the first open position.

[0090] As illustrated in FIG. **9**, the recording section **48** includes the carriage **51** and a liquid ejection head **52**. The carriage **51** supports the liquid ejection head **52** from the upward direction **Z1**. The liquid ejection head **52** is provided in the downward direction **22** from the carriage **51**. That is, the liquid ejection head **52** is mounted on the carriage **51**.

[0091] The liquid ejection head **52** includes a plurality of nozzles (not shown). Each of the plurality of nozzles is opened toward the medium supported by a support section **53** (to be described later). Each of the plurality of nozzles ejects a liquid. The liquid ejection head **52** ejects liquid from the plurality of nozzles toward the medium supported by the support section **53**. As described above, the recording section **48** is configured to perform recording on a medium by ejecting liquid onto the medium. The liquid ejection head **52** is a serial head type which ejects liquid in accordance with the movement of the carriage **51** in the widthwise direction X but may be a line head type.

[0092] The recording device **11** includes the support section **53**. The support section **53** is mounted on the main body **21**. The support section **53** is configured to support a medium. In particular, the support section **53** supports the medium in a recording region RA in which recording is performed on the medium by the recording section **48**.

[0093] The recording device **11** includes a medium placement section **54**. One or several media can be placed on the medium placement section **54**. The medium placement section **54** may be a tray on which a medium can be placed. The medium placement section **54** is mounted on the main body **21**. The medium placement section **54** can be displaced from the main body **21** along the front-rear direction Y. By being pulled out from the main body **21** to the frontward direction **Y1**, the medium placement section **54** is in a state in which medium can be placed thereon or taken out therefrom. In this way, the frontward direction **Y1** corresponds to an example of a pull-out direction in which the medium placement section **54** is pulled out.

[0094] The medium placement section **54** includes a partition **54A**. The partition **54A** is positioned at the center of the medium placement section **54** in the front-rear direction Y. The partition **54A** partitions the medium placement section **54** in the front-rear direction Y. The medium placement section **54** includes a first medium placement section **54B**. The medium placement section **54** may include a second medium placement section **54C**. The first medium placement section **54B** and the second medium placement section **54C** are regions partitioned in the front-rear direction Y by the partition **54A**. The first medium placement section **54B** is positioned further in the backward direction **Y2** than the second medium placement section **54C**. The first medium placement section **54B** can accommodate one or more media before reading. One or more media after reading can be placed on the second medium placement section **54C**.

[0095] The recording device **11** includes a medium feeding section **59**. The medium feeding section **59** is positioned in the upward direction **Z1** of the medium placement section **54** when the medium placement section **54** is mounted to the main body **21**. More specifically, the medium feeding section **59** is positioned in the upward direction **Z1** of the first medium placement section **54B** when the medium placement section **54** is installed in the main body **21**. The medium feeding section **59** feeds the medium placed on the medium placement section **54** one by one.

[0096] The recording device **11** includes a transport section **55**. The transport section **55** is configured to transport a medium. The transport section **55** is mounted on the main body **21**. The transport section **55** transports the medium fed by the medium feeding section **59** along the transport path **49**.

[0097] The transport section **55** includes a plurality of rollers and plurality of motors serving as driving sources. In particular, the transport section **55** includes a transport roller pair **56**. The transport roller pair **56** is positioned upstream of the transport path **49** of the support section **53**. The transport roller pair **56** transports the medium to the support section **53** along the transport path **49**. The transport section **55** includes a discharge roller pair **57**. The discharge roller pair **57** is positioned downstream of the transport path **49** of the support section **53**. The discharge roller pair **57** discharges the medium after recording along the transport path **49**.

[0098] In this way, the medium placed on the medium placement section **54** is fed by the medium feeding section **59**. Then, the medium fed by the medium feeding section **59** is transported along the transport path **49** by the transport section **55**. By this, the recording section **48** performs recording on the medium transported along the transport path **49** by the transport section **55**. That is, the recording section **48** performs recording on the medium fed by the medium feeding section **59**.

[0099] The transport section **55** includes an inversion section **58**. That is, the inversion section **58** is mounted on the main body **21**. In addition, the recording device **11** includes the inversion section **58**. The inversion section **58** may be configured to be attachable to and detachable from the main body **21** from the backward direction **Y2**. The inversion section **58** includes the plurality of rollers. The inversion section **58** is configured to transport a medium along the transport path **49**.

[0100] The transport path **49** includes an inversion path **49A**. The inversion path **49A** is a path coupling the transport roller pair **56** and the inversion section **58**. The inversion path **49A** extends along the front-rear direction **Y**. After recording is performed on the front surface of the medium by the recording section **48** in the recording region **RA**, the transport roller pair **56** transports the medium to the inversion section **58** along the inversion path **49A**. The inversion section **58** is configured to transport the medium along the transport path **49** in a state where the medium transported along the inversion path **49A** is inverted. That is, the inversion section **58** inverts the medium fed by the medium feeding section **59**.

[0101] In this way, the medium inverted by the inversion section **58** is transported to the recording region **RA** again, and thus the recording section **48** can perform recording on the back surface of the medium. By this, the recording section **48** can print on both sides of the medium. That is, the recording section **48** can perform recording after the medium is inverted by the inversion section **58**. In particular, the recording section **48** may be able to record on double-sided printable photo paper sheet as a medium.

[0102] The inversion section **58** is detachably attached on the main body **21**. The inversion section **58** can be removed from the backward direction **Y2**. The inversion section **58** may be removable from the backward direction **Y2** when the main body **21** is positioned in the closed position. When the inversion section **58** is detached, the medium jammed in the inversion section **58** can be removed.

Access to Recording Device **11**

[0103] As shown in FIGS. **6** to **9**, the plurality of recording devices **11** can be housed in the first accommodation section **71** of the carry case **100** in a state of being stacked in the vertical direction **Z**. The carry case **100** is configured to facilitate access to the plurality of recording devices **11** when the first opening and closing door **61** and the second opening and closing door **62** are in an open state.

[0104] Specifically, in a state where the first opening and closing door **61** is open, the operation section **34** can be operated in the plurality of recording devices **11** housed in the first accommodation section **71**. In a state where the first opening and closing door **61** is open, the display section **35** in the plurality of recording devices **11** housed in the first accommodation section **71** can be visually checked. In a state where the first opening and closing door **61** is open, the opening and closing section **37** can be opened and closed. In a state where the first opening and closing door **61** is open, the sheet discharge tray **37B** can expand and contract in the front-rear

direction Y.

[0105] In a state where the first opening and closing door **61** is open, it is possible to perform an operation of pulling out the main bodies **21** of the plurality of recording devices **11** housed in the first accommodation section **71**. By this, it is possible to perform a maintenance operation on the plurality of recording devices **11** housed in the first accommodation section **71**.

[0106] As a specific example, as a maintenance operation, a pouring operation in which liquid is poured into the liquid container **40** can be performed. In a case where the liquid container **40** is configured to be attachable and detachable, an operation of attaching and detaching the liquid container **40** can be performed as the maintenance operation. As the maintenance operation, the attaching and detaching operation of the waste liquid container **46** can be performed. As the maintenance operation, for example, a medium that is jammed around the vicinity or the like of the recording section **48** in the transport path **49** can be removed.

[0107] In a state where the second opening and closing door **62** is open, the plurality of recording devices **11** housed in the first accommodation section **71** can be operated. As a specific example, an operation of coupling a power supply cable to a power supply terminal provided on the rear surface **13** of the recording device **11** can be performed.

[0108] In a state where the second opening and closing door **62** is open, maintenance operations on a plurality of recording devices **11** housed in the first accommodation section **71** can be performed. As a specific example, the inversion section **58** can be removed from the main body **21**. By this, a medium that is jammed around the vicinity or the like of the inversion section **58** in the transport path **49** can be removed.

Operation and Effects of First Embodiment

[0109] Operations and effects of the first embodiment will be described.

[0110] (1) In the carry case **100**, the plurality of recording devices **11** can be housed in the first accommodation section **71** in a state of being stacked in plurality of tiers in the vertical direction Z, and the power supply device **91** can be housed in the second accommodation section **72**. By this, the carry case **100** in which the plurality of recording devices **11** and the power supply device **91** are housed, using the wheels **63**, can be transported. Therefore, it is possible to improve user convenience.

[0111] (2) In the carry case **100**, electric power from the power supply device **91** housed in the second accommodation section **72** can be supplied to the plurality of recording devices **11** housed in the first accommodation section **71** in a state of being stacked in the plurality of tiers in the vertical direction Z. By this, by transporting the carry case **100**, an environment, in which the plurality of recording devices **11** are used, can be arranged. Therefore, it is possible to improve user convenience.

[0112] (3) A first opening and closing door **61** is provided on the case front surface **100A** that can be opened and closed. A second opening and closing door **62** is provided on the case rear surface **100B**, which can be opened and closed. By this, while the first opening section **60A** and the second opening section **60B** of the housing main body **60** can be closed, access to the inside of the housing main body **60** can be facilitated. Therefore, it is possible to improve user convenience.

[0113] (4) The first accommodation region **71B** is arranged further in an upward direction **Z1** than the second accommodation region **72B**. By this, it is possible to improve operability with respect to the plurality of recording devices **11** housed in the first accommodation region **71B**. Therefore, it is possible to improve user convenience.

[0114] (5) By expanding the table **65** from storage position **P11** to expansion position **P12**, the table **65** can be used in various ways. Therefore, it is possible to improve user convenience.

[0115] (6) When the table **65** is expanded from the storage position **P11** to the expansion position **P12**, the table **65** can be supported by the opened the first opening and closing door **61** and the second opening and closing door **62**. By this, the table **65** can be used stably. Therefore, it is possible to improve user convenience.

[0116] (7) When the table **65** is expanded from the storage position **P11** to the expansion position **P12**, the object to be placed can be placed on the second placement surface **65B**. Therefore, it is possible to improve user convenience.

[0117] (8) By adjusting the height using the adjuster **66**, the first opening and closing door **61** and the second opening and closing door **62** can be supported from the downward direction **Z2**. By this, the table **65** can be used stably. Therefore, it is possible to improve user convenience.

[0118] (9) When the user grips the grip section **64**, the operability of the carry case **100** can be improved. In particular, the housing main body **60** can be easily inclined by gripping the grip section **64**, and the operability of the carry case **100** can be improved. Therefore, it is possible to improve user convenience.

[0119] (10) By opening the first opening and closing door **61**, it is possible to perform maintenance operations on the plurality of recording devices **11** housed in the first accommodation section **71** from the case front surface **100A** side. Therefore, it is possible to improve user convenience.

[0120] (11) By opening the second opening and closing door **62**, it is possible to perform maintenance operations on the plurality of recording devices **11** housed in the first accommodation section **71** from the case rear surface **100B** side. Therefore, it is possible to improve user convenience.

[0121] (12) By opening the first opening and closing door **61**, the operation of pulling out the main bodies **21** of the plurality of recording devices **11** housed in the first accommodation section **71** can be performed from the case front surface **100A** side. Therefore, it is possible to improve user convenience.

[0122] (13) The first opening and closing door **61** can be locked by the first locking section **67A**. The second opening and closing door **62** can be locked by the second locking section **67B**. By this, the security of the carry case **100** can be improved. Therefore, it is possible to improve user convenience.

[0123] (14) The illumination section **68** illuminates the inside of the housing main body **60**. Therefore, the visual checking ability about inside of the housing main body **60** can be improved. Therefore, it is possible to improve user convenience.

[0124] (15) The wheels **63** can be driven by the driving force from the drive section **69** housed in the second accommodation section **72**. By this, the operability of the carry case **100** can be improved. Therefore, it is possible to improve user convenience.

[0125] (16) Even in a case where the housing main body **60** is inclined at a predetermined angle, it is possible to horizontally maintain the plurality of recording devices **11** which are housed in the first accommodation section **71**. By this, it is possible to suppress the liquid from leaking from at least one or more the plurality of recording devices **11**. Therefore, it is possible to improve user convenience.

[0126] (17) In a case where the plurality of recording devices **11** is mounted in a state of being stacked in the plurality of tiers in the vertical direction **Z**, the plurality of recording devices **11** can be maintained horizontally by the guide section **74** guiding the first placement section **71A**. By this, it is possible to suppress the liquid from leaking from at least one or more the plurality of recording devices **11**. Therefore, it is possible to improve user convenience.

[0127] (18) At the time of recording on medium, in a case where the housing main body **60** is arranged horizontally, the first placement section **71A** is fixed to the guide section **74** at a position where the horizontal is maintained by the second fixing section **75**. By this, it is possible to suppress a decrease in recording accuracy on the medium. Therefore, it is possible to improve user convenience.

[0128] (19) The distances between the guide section **74** and the first placement section **71A** can be maintained within a permissible range. By this, even when vibration is applied to the carry case **100**, it is possible to prevent the first placement section **71A** from being separated from the guide section **74**. Therefore, the operability of the carry case **100** can be improved. Therefore, it is

possible to improve user convenience.

MODIFIED EXAMPLES

[0129] The present embodiment can be implemented with the following modifications. The present embodiment and the following modified examples can be implemented in combination with each other as long as there is no technical contradiction. [0130] The second fixing section **75** may be provided at any position. The second fixing section **75** may be configured to be insertable and removable by a user's operation on the top surface of the first placement section **71A**. The second fixing section **75** may be insertable and removable along the widthwise direction **X**, or may be insertable and removable along the front-rear direction **Y**. The second fixing section **75** may fix the plurality of recording devices **11** to the first placement section **71A** by coming into contact with the plurality of recording devices **11** from the upward direction **Z1**. In such a case, the first placement section **71A** is fixed to the guide section **74**. [0131] The carry case **100** may include a biasing section that biases the plurality of recording devices **11** placed on the first placement section **71A** from the upward direction **Z1** side to the first placement section **71A**. In this way, the first placement section **71A** is biased toward the guide section **74** by the biasing force from the biasing section. By this, the distance between the guide section **74** and the first placement section **71A** is maintained within the permissible range. [0132] At least one of the surfaces of the guide groove **74A** and the rotator **71C** may be formed of an elastic component so as to absorb vibration in the vertical direction **Z**. More specifically, the surface of the guide groove **74A** and the surface of the rotator **71C** may be formed of an elastic material such as rubber. [0133] The housing main body **60** may include a housing space for housing the wheels **63** on the downward direction **Z2** side. That is, a part of the wheels **63** may be housed in the housing space of the housing main body **60**. [0134] The first opening and closing door **61** and the second opening and closing door **62** may include a window section through which the inside of the carry case **100** can be visually checked from the outside. The window section is translucent. By this, the states of the recording device **11** and the power supply device **91** housed in the carry case **100** can be visually checked from the outside of the carry case **100** through the window section. In particular, the display content of the display section **35** of the recording device **11** housed in the carry case **100** can be visually checked from the outside of the carry case **100** through the window section. [0135] The table **65** may be slidable between the storage position **P11** and the expansion position **P12**. The table **65** may include a support member for being supported by the installation surface **99**. In this case, the table **65** may not be supported by the opened first opening and closing door **61** and the opened second opening and closing door **62**. [0136] The first opening and closing door **61** may include the plurality of adjusters **66**. The second opening and closing door **62** may include the plurality of adjusters **66**. The carry case **100** may not include the adjuster **66**. [0137] The first opening and closing door **61** and the second opening and closing door **62** may be biparting doors on the first widthwise direction **X1** side and the second widthwise direction **X2** side of the housing main body **60**. The first opening and closing door **61** and the second opening and closing door **62** may be openable and closable by a rotation shaft along the widthwise direction **X**. The first opening and closing door **61** and the second opening and closing door **62** may be sliding doors. The first opening and closing door **61** and the second opening and closing door **62** may be able to place the object to be placed to be placed when opened. [0138] The carry case **100** may include a third accommodation section in addition to the first accommodation section **71** and the second accommodation section **72**. The third accommodation section may be provided between the first accommodation section **71** and the second accommodation section **72** in the vertical direction **Z**. The third accommodation section may be provided in a downward direction **Z2** than the second accommodation section **72** or may be provided in an upward direction **Z1** than the first accommodation section **71**. The third accommodation section may be provided so as to be aligned with the second accommodation section **72** in the widthwise direction **X**. The first accommodation section **71** may be positioned in a downward direction **Z2** lower than the second accommodation section **72**. [0139] The second

accommodation section **72** may be able to house a component other than the power supply device **91**. The second accommodation section **72** may be able to accommodate maintenance supplies for performing maintenance. As a specific example, the second accommodation section **72** may be able to house a liquid accommodation body such as an ink bottle or a replacement ink cartridge. The second accommodation section **72** may be able to house a waste liquid container such as a replacement waste liquid tank. [0140] The carry case **100** may be constituted of any material. As a specific example, the carry case **100** may be made of, for example, aluminum, duralumin, polycarbonate, or polypropylene, or may be made of cloth and a core material. [0141] A lateral type printer may be adopted as the recording device **11**. The lateral type printer is a printer in which carriage **51** can move in two directions of the main scanning direction and the sub-scanning direction. [0142] The medium may be any medium, and may be, for example, paper sheet, a resin film or sheet, a composite film of a resin and a metal, a laminate film, roll paper, a woven fabric, a nonwoven fabric, a metal foil, a metal film, a ceramic sheet, or clothing. [0143] The liquid can be arbitrarily selected as long as it can adhere to the medium to perform recording on the medium. For example, the ink includes an ink in which particles of a functional material made of a solid material such as a pigment or metal particles are dissolved, dispersed, or mixed in a solvent and includes various compositions such as a water-based ink, an oil-based ink, a gel ink, and a hot-melt ink. [0144] As used herein, the phrase “at least one of” means one or more of the desired options. As an example, the phrase “at least one of” as used herein means only one option if the number of options is two, or both of the two options. As another example, the phrase “at least one of” as used herein means only one option or any combination of two or more options, as long as the number of options is three or more.

NOTES

[0145] Hereinafter, technical ideas grasped from the above-described embodiment and modified examples, and operational effects thereof will be described. The present technical idea and the operational effects thereof can be combined with each other within a technically consistent range.

[0146] (A1) A carry case includes a housing main body; a first accommodation section configured to accommodate, in a first accommodation region within the housing main body, a plurality of recording devices in a state of being stacked in a plurality of tiers in a vertical direction; a second accommodation section configured to accommodate, in the housing main body in a second accommodation region that is aligned in the vertical direction with the first accommodation region, a power supply device that supplies electric power to the plurality of recording devices; a first opening and closing door configured to open and close in a first surface of the housing main body; a second opening and closing door configured to open and close in a second surface opposite to the first surface of the housing main body; and a wheel that is below the housing main body and that enables moving the housing main body.

[0147] According to this configuration, the carry case can house the plurality of recording devices in a state of being stacked in plurality of tiers in the vertical direction in the first accommodation section and can house the power supply device in the second accommodation section. By this, the carry case housing the plurality of recording devices and the power supply device can be transported using the wheels. Therefore, it is possible to improve user convenience.

[0148] In the carry case, it is possible to supply electric power from the power supply device housed in the second accommodation section to the plurality of recording devices housed in the first accommodation section in a state of being stacked in the plurality of tiers in the vertical direction. By this, it is possible to prepare an environment in which plurality of recording devices are used by transporting the carry case. Therefore, it is possible to improve user convenience.

[0149] A first opening and closing door is provided to be openable and closable on the first surface, and a second opening and closing door is provided to be openable and closable on the second surface. By this, the inside of the housing main body can be shielded, and access to the inside of the housing main body can be facilitated. Therefore, it is possible to improve user convenience.

[0150] (A2) The carry case may further include a table provided above the housing main body, wherein the table is expandable from a storage position to an expansion position and, when the table is expanded to the expansion position, may be supported by the opened first opening and closing door and the opened second opening and closing door.

[0151] According to this configuration, it is possible to diversify the use of the table by expanding it from the storage position to the expansion position. Therefore, it is possible to improve user convenience.

[0152] Further, when the table is expanded from the storage position to the expansion position, the table can be supported by the opened first opening and closing door and the opened second opening and closing door. By this, it is possible to stably use the table. Therefore, it is possible to improve user convenience.

[0153] (A3) In the above carry case, the table may have a placement surface on which an object to be placed can be placed when the table is expanded to the expansion position.

[0154] According to this configuration, when the table is expanded from the storage position to the expansion position, the object to be placed can be placed on the placement surface. Therefore, it is possible to improve user convenience.

[0155] (A4) The carry case may include an adjuster configured to, when the table is expanded to the expansion position, adjust the height below the first opening and closing door and below the second opening and closing door.

[0156] According to this configuration, the adjustment of the height of the adjuster can support the first opening and closing door and the second opening and closing door from the downward direction. By this, it is possible to stably use the table. Therefore, it is possible to improve user convenience.

[0157] (A5) The carry case may include a grip section that can be gripped by a user.

[0158] According to this configuration, since the grip section is gripped by the user, it is possible to improve the operability of the carry case. Therefore, it is possible to improve user convenience.

[0159] (A6) In the carry case, the first opening and closing door may be configured to open and close at least a part of the first accommodation region and in a state where the first opening and closing door is open, maintenance operations may be performable on the plurality of recording devices accommodated in the first accommodation section.

[0160] According to this configuration, by opening the first opening and closing door, it is possible to perform maintenance operations on the plurality of recording devices housed in the first accommodation section from the first surface side. Therefore, it is possible to improve user convenience.

[0161] (A7) In the carry case, the second opening and closing door is configured to open and close at least a part of the first accommodation region, and in a state where the second opening and closing door is open, maintenance operations may be performable on the plurality of recording devices accommodated in the first accommodation section.

[0162] According to this configuration, by opening the second opening and closing door, it is possible to perform maintenance operations on the plurality of recording devices housed in the first accommodation section from the second surface side. Therefore, it is possible to improve user convenience.

[0163] (A8) In the carry case, the first opening and closing door is configured to open and close at least a part of the first accommodation region and the main bodies of the plurality of recording devices housed in the first accommodation section may be configured to be pulled out in a state where the first opening and closing door is open.

[0164] According to this configuration, by opening the first opening and closing door, it is possible to perform an operation of pulling out the main bodies of the plurality of recording devices housed in the first accommodation section from the first surface side. Therefore, it is possible to improve user convenience.

[0165] The carry case may include a first locking section configured to lock the first opening and closing door in a state in which the first opening and closing door is closed and a second locking section configured to lock the second opening and closing door in a state in which the second opening and closing door is closed.

[0166] According to this configuration, the first opening and closing door and the second opening and closing door can be locked. By this, it is possible to improve the security of the carry case. Therefore, it is possible to improve user convenience.

[0167] (A10) The carry case may include an illumination section configured to illuminate the inside of the housing main body.

[0168] According to this configuration, the visual checking ability about inside of the housing main body can be improved. Therefore, it is possible to improve user convenience.

[0169] (A11) The carry case may include a drive section configured to drive the wheels, wherein the drive section may be configured to be accommodated in the second accommodation section.

[0170] According to this configuration, the wheels can be driven by housing the drive section in the second accommodation section. By this, the operability of the carry case can be improved. Therefore, it is possible to improve user convenience.

[0171] The recording system includes a plurality of recording devices; a power supply device configured to supply power to the plurality of recording devices; and a carry case configured to accommodate the plurality of recording devices and the power supply device, wherein the carry case has a housing main body; a first accommodation section configured to accommodate the plurality of recording devices in a state of being stacked in a plurality of tiers in a vertical direction in the first accommodation region in the housing main body, a second accommodation section configured to house the power supply device in a second accommodation region aligned with the first accommodation region in the vertical direction in the housing main body, a first opening and closing door configured to open and close in a first surface of the housing main body; a second opening and closing door configured to open and close in a second surface opposite to the first surface of the housing main body; and a wheel that is below the housing main body and that enables moving the housing main body. According to this configuration, the same effect as that of the A1 can be achieved.

[0172] (B1) A carry case includes a housing main body; a first accommodation section configured to accommodate a plurality of recording devices in a state of being stacked in a plurality of tiers in a vertical direction in a first accommodation region within the housing main body, the recording devices performing recording by ejecting liquid; a second accommodation section configured to accommodate, in the housing main body in a second accommodation region that is aligned in the vertical direction with the first accommodation region, a power supply device that supplies electric power to the plurality of recording devices; a wheel that is below the housing main body and that enables moving the housing main body; and a horizontal maintaining section configured to, when the housing main body is inclined at a predetermined angle, horizontally support the plurality of recording devices accommodated in the first accommodation section.

[0173] According to this configuration, it can house the plurality of recording devices in a state of being stacked in plurality of tiers in the vertical direction in the first accommodation section and can house the power supply device in the second accommodation section. By this, the carry case housing the plurality of recording devices and the power supply device can be transported using the wheels. Therefore, it is possible to improve user convenience.

[0174] In addition, it is possible to supply electric power from the power supply device housed in the second accommodation section to the plurality of recording devices housed in the first accommodation section in a state of being stacked in plurality of tiers in the vertical direction. By this, it is possible to prepare an environment in which plurality of recording devices are used by transporting the carry case. Therefore, it is possible to improve user convenience.

[0175] Even when the housing main body is inclined at a predetermined angle, the plurality of

recording devices accommodated in the first accommodation section can be maintained horizontally. By this, it is possible to suppress the liquid from leaking from at least one or more the plurality of recording devices. Therefore, it is possible to improve user convenience.

[0176] (B2) In the carry case, the horizontal maintaining section, when the housing main body is inclined at a predetermined angle, does not need to horizontally support the power supply device housed in the second accommodation section.

[0177] According to this configuration, when the housing main body is inclined at the predetermined angle, the power supply device housed in the second accommodation section is not supported horizontally. By this, it possible to reduce the size and weight of the carry case. Therefore, the carry case can be easily transported. Therefore, it is possible to improve user convenience.

[0178] (B3) In the carry case, the second accommodation section may include a second placement section that places the power supply device and a first fixing section that fixes the position of the power supply device placed on the second placement section.

[0179] According to this configuration, when the housing main body is inclined at the predetermined angle, the power supply device housed in the second accommodation section is not supported horizontally. However, the power supply device can be fixed to the second placement section by the first fixing section. By this, the power supply device can be stably transported. Therefore, the carry case can be easily transported. Therefore, it is possible to improve user convenience.

[0180] (B4) The carry case may further include a first opening and closing door configured to open and close in a first surface of the housing main body and a second opening and closing door configured to open and close in a second surface opposite to the first surface of the housing main body.

[0181] According to this configuration, the first opening and closing door is provided on the first surface so as to be openable and closable, and the second opening and closing door is provided on the second surface so as to be openable and closable. By this, the inside of the housing main body can be shielded, and access to the inside of the housing main body can be facilitated. Therefore, it is possible to improve user convenience.

[0182] (B5) In the carry case, the first opening and closing door is configured to open and close at least a part of the first accommodation region and in a state where the first opening and closing door is open, maintenance operations may be performable on the plurality of recording devices accommodated in the first accommodation section.

[0183] According to this configuration, by opening the first opening and closing door, it is possible to perform maintenance operations on the plurality of recording devices housed in the first accommodation section from the first surface side. Therefore, it is possible to improve user convenience.

[0184] (B6) In the carry case, the second opening and closing door is configured to open and close at least a part of the first accommodation region, and in a state where the second opening and closing door is open, maintenance operations may be performable on the plurality of recording devices accommodated in the first accommodation section.

[0185] According to this configuration, by opening the second opening and closing door, it is possible to perform maintenance operations on the plurality of recording devices housed in the first accommodation section from the second surface side. Therefore, it is possible to improve user convenience.

[0186] (B7) In the carry case, the first opening and closing door is configured to open and close at least a part of the first accommodation region and the main bodies of the plurality of recording devices housed in the first accommodation section may be configured to be pulled out in a state where the first opening and closing door is open.

[0187] According to this configuration, by opening the first opening and closing door, it is possible

to perform an operation of pulling out the main bodies of the plurality of recording devices housed in the first accommodation section from the first surface side. Therefore, it is possible to improve user convenience.

[0188] (B8) The carry case may further include a first locking section configured to lock the first opening and closing door in a state in which the first opening and closing door is closed and a second locking section configured to lock the second opening and closing door in a state in which the second opening and closing door is closed.

[0189] According to this configuration, the first opening and closing door and the second opening and closing door can be locked. By this, it is possible to improve the security of the carry case. Therefore, it is possible to improve user convenience.

[0190] (B9) The carry case may further include a table provided above the housing main body, wherein the table is expandable from a storage position to an expansion position and, when the table is expanded to the expansion position, may be supported by the opened first opening and closing door and the opened second opening and closing door.

[0191] According to this configuration, it is possible to diversify the use of the table by expanding it from the storage position to the expansion position. Therefore, it is possible to improve user convenience.

[0192] Further, when the table is expanded from the storage position to the expansion position, the table can be supported by the opened first opening and closing door and the opened second opening and closing door. By this, it is possible to stably use the table. Therefore, it is possible to improve user convenience.

[0193] (B10) In the above carry case, the table may have a placement surface on which an object to be placed is placeable when the table is expanded to the expansion position.

[0194] According to this configuration, when the table is expanded from the storage position to the expansion position, the object to be placed can be placed on the placement surface. Therefore, it is possible to improve user convenience.

[0195] (B11) The carry case may further include an adjuster configured to, when the table is expanded to the expansion position, adjust the height below the first opening and closing door and below the second opening and closing door.

[0196] According to this configuration, the adjustment of the height of the adjuster can support the first opening and closing door and the second opening and closing door from the downward direction. By this, it is possible to stably use the table. Therefore, it is possible to improve user convenience.

[0197] (B12) The carry case may further include a grip section that can be gripped by a user.

[0198] According to this configuration, the user can easily transport the carry case by holding the grip section. Therefore, it is possible to improve user convenience.

[0199] (B13) The carry case may further include an illumination section configured to illuminate the inside of the housing main body.

[0200] According to this configuration, the visual checking ability about inside of the housing main body can be improved. Therefore, it is possible to improve user convenience.

[0201] (B14) The carry case may further include a drive section configured to drive the wheels, wherein the drive section may be configured to be accommodated in the second accommodation section.

[0202] According to this configuration, the wheels can be driven by housing the drive section in the second accommodation section. By this, the carry case can be easily transported. Therefore, it is possible to improve user convenience.

[0203] (B15) In the carry case, the first accommodation section includes a first placement section on which the plurality of recording devices are placeable in a state of being stacked in plurality of tiers in the vertical direction, the horizontal maintaining section includes a guide section that guides the first placement section and a second fixing section that fixes the first placement section to the

guide section, the guide section is fixed to the housing main body and, when the housing main body is inclined at a predetermined angle, guides the first placement section to keep the first placement section horizontal, and when the housing main body is arranged horizontally, the second fixing section may fix the first placement section to the guide section at a position at which the first placement section is kept horizontal.

[0204] According to this configuration, it is possible to horizontally maintain the plurality of recording devices by guiding the first placement section in a state in which the plurality of recording devices is mounted in a state of being stacked in plurality of tiers in the vertical direction. By this, it is possible to suppress the liquid from leaking from at least one or more the plurality of recording devices. Therefore, it is possible to improve user convenience.

[0205] When the housing main body is horizontally arranged at the time of recording on the medium, the first placement section is fixed to the guide section at a position at which the first placement section is maintained to be horizontal. By this, it is possible to suppress a decrease in recording accuracy on the medium. Therefore, it is possible to improve user convenience.

[0206] (B16) In the carry case, the first accommodation section includes a first placement section on which the plurality of recording devices are placeable in a state of being stacked in plurality of tiers in the vertical direction, the horizontal maintaining section includes a guide section that guides the first placement section, the guide section is fixed to the housing main body and, when the housing main body is inclined at a predetermined angle, guides the first placement section to keep the first placement section horizontal, and the carry case may be configured so as to maintain a distance between the guide section and the first placement section within an permissible range.

[0207] According to this configuration, it is possible to horizontally maintain the plurality of recording devices by guiding the first placement section in a state in which the plurality of recording devices is mounted in a state of being stacked in plurality of tiers in the vertical direction. By this, it is possible to suppress the liquid from leaking from at least one or more the plurality of recording devices. Therefore, it is possible to improve user convenience.

[0208] The distance between the guide section and the first placement section can be maintained within an permissible range. By this, even when vibration is applied to the carry case, it is possible to suppress the first placement section from being separated from the guide section. Therefore, the carry case can be easily transported.

[0209] Therefore, it is possible to improve user convenience.

[0210] (B17) The recording system includes a plurality of recording devices for performing printing by ejecting liquid; a power supply device configured to supply power to the plurality of recording devices; and a carry case configured to accommodate the plurality of recording devices and the power supply device; wherein the carry case has a housing main body, a first accommodation section configured to accommodate the plurality of recording devices in a state of being stacked in a plurality of tiers in a vertical direction in the first accommodation region in the housing main body, a second accommodation section configured to house the power supply device in a second accommodation region aligned with the first accommodation region in the vertical direction in the housing main body, a wheel that is below the housing main body and that enables moving the housing main body; and a horizontal maintaining section configured to, when the housing main body is inclined at a predetermined angle, horizontally support the plurality of recording devices accommodated in the first accommodation section. According to this configuration, the same effect as that of B1 can be achieved.

Claims

1. A carry case comprising: a housing main body; a first accommodation section configured to accommodate, in a first accommodation region within the housing main body, a plurality of recording devices in a state of being stacked in a plurality of tiers in a vertical direction; a second

accommodation section configured to accommodate, in the housing main body in a second accommodation region that is aligned in the vertical direction with the first accommodation region, a power supply device that supplies electric power to the plurality of recording devices; a first opening and closing door configured to open and close in a first surface; a second opening and closing door configured to open and close in a second surface that is opposite from the first surface; and a wheel that is below the housing main body and that enables moving the housing main body.

2. The carry case according to claim 1, further comprising: a table provided above the housing main body, wherein the table is expandable from a storage position to an expansion position and, when the table is expanded to the expansion position, is supported by the opened first opening and closing door and the opened second opening and closing door.

3. The carry case according to claim 2, wherein the table has a placement surface on which an object to be placed is placeable when the table is expanded to the expansion position.

4. The carry case according to claim 3, further comprising: an adjuster configured to, when the table is expanded to the expansion position, adjust the height below the first opening and closing door and below the second opening and closing door.

5. The carry case according to claim 1, further comprising: a grip section configured to be gripped by a user.

6. The carry case according to claim 1, wherein the first opening and closing door is configured to open and close at least a part of the first accommodation region and in a state where the first opening and closing door is open, maintenance operations are performable on the plurality of recording devices accommodated in the first accommodation section.

7. The carry case according to claim 1, wherein the second opening and closing door is configured to open and close at least a part of the first accommodation region, and in a state where the second opening and closing door is open, maintenance operations are performable on the plurality of recording devices accommodated in the first accommodation section.

8. The carry case according to claim 1, wherein the first opening and closing door is configured to open and close at least a part of the first accommodation region and the main bodies of the plurality of recording devices housed in the first accommodation section are configured to be pulled out in a state where the first opening and closing door is open.

9. The carry case according to claim 1, further comprising: a first locking section configured to lock the first opening and closing door in a state in which the first opening and closing door is closed and a second locking section configured to lock the second opening and closing door in a state in which the second opening and closing door is closed.

10. The carry case according to claim 1, further comprising: an illumination section configured to illuminate the inside of the housing main body.

11. The carry case according to claim 1, further comprising: a drive section configured to drive the wheel, wherein the drive section is configured to be accommodated in the second accommodation section.

12. A carry case comprising: a housing main body; a first accommodation section configured to accommodate a plurality of recording devices in a state of being stacked in a plurality of tiers in a vertical direction in a first accommodation region within the housing main body, the recording devices performing recording by ejecting liquid; a second accommodation section configured to accommodate, in the housing main body in a second accommodation region that is aligned in the vertical direction with the first accommodation region, a power supply device that supplies electric power to the plurality of recording devices; a wheel that is below the housing main body and that enables moving the housing main body; and a horizontal maintaining section configured to, when the housing main body is inclined at a predetermined angle, horizontally support the plurality of recording devices accommodated in the first accommodation section.

13. The carry case according to claim 12, wherein when the housing main body is inclined at a predetermined angle, the horizontal maintaining section does not horizontally support the power

supply device housed in the second accommodation section.

14. The carry case according to claim 12, wherein the second accommodation section includes a second placement section on which the power supply device is placeable and a first fixing section that fixes the position of the power supply device placed on the second placement section.

15. The carry case according to claim 12, wherein the first accommodation section includes a first placement section on which the plurality of recording devices are placeable in a state of being stacked in plurality of tiers in the vertical direction, the horizontal maintaining section includes a guide section that guides the first placement section and a second fixing section that fixes the first placement section to the guide section, the guide section is fixed to the housing main body and, when the housing main body is inclined at a predetermined angle, guides the first placement section to keep the first placement section horizontal, and when the housing main body is arranged horizontally, the second fixing section fixes the first placement section to the guide section at a position at which the first placement section is kept horizontal.

16. The carry case according to claim 12, wherein the first accommodation section includes a first placement section on which the plurality of recording devices are placeable in a state of being stacked in plurality of tiers in the vertical direction, the horizontal maintaining section includes a guide section that guides the first placement section, the guide section is fixed to the housing main body and, when the housing main body is inclined at a predetermined angle, guides the first placement section to keep the first placement section horizontal, and the carry case is configured so as to maintain a distance between the guide section and the first placement section within an permissible range.

17. A recording system comprising: a plurality of recording devices; a power supply device configured to supply power to the plurality of recording devices; a carry case configured to accommodate the plurality of recording devices and the power supply device, wherein the carry case includes a housing main body, a first accommodation section configured to accommodate the plurality of recording devices in a state of being stacked in a plurality of tiers in a vertical direction in the first accommodation region in the housing main body, a second accommodation section configured to house the power supply device in a second accommodation region aligned with the first accommodation region in the vertical direction in the housing main body, a first opening and closing door configured to open and close in a first surface, a second opening and closing door configured to open and close in a second surface opposite to the first surface, and a wheel that is below the housing main body and that enables moving the housing main body.
